PHYSICS

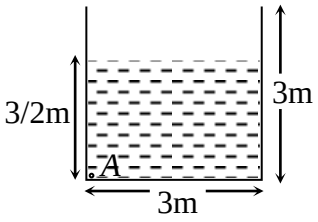
SECTION – I

Single Correct Choice Type

This section contains 8 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

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1. A rectangular container of height and width 3m each is filled with an ideal liquid upto a height of $\frac{3}{2}$m. This container is pulled horizontally by an acceleration equal to g. The gauge pressure at point A as shown in figure is P_1. If the container is completely filled with the liquid and closed from top and then pulled horizontally by same acceleration, the gauge pressure at same point is P_2. Then $\frac{P_1}{P_2}$ is



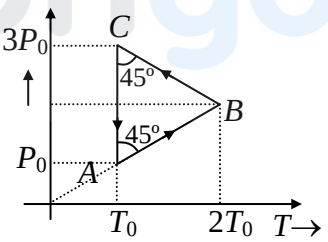
(a) 1 (b) $\frac{1}{3}$ (c) $\frac{1}{4}$ (d) $\frac{1}{2}$

2. A steel wire with cross section 3 cm^2 has elastic limit 2.4×10^8pa. The maximum upward acceleration that can be given to a 1200 kg elevator supported by this cable if the stress is not to exceed 1/3rd of the elastic limit is

(a) $9ms^{-2}$ (b) $10ms^{-2}$ (c) $11ms^{-2}$ (d) $12ms^{-2}$

3. An ideal monoatomic gas is undergoing a cyclic process as shown in the figure. Then choose the correct statement

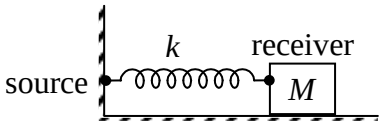
(a) work done by the gas in process CA is negative
(b) heat is rejected by the system in process BC
(c) work done by the gas in the cyclic process is positive
(d) internal energy of the gas decreases in process AB



4. A satellite is projected with velocity $\sqrt{\frac{5}{6}}$ times its escape speed from earth's surface. The initial velocity of the satellite is parallel to the surface of earth. The maximum distance of the satellite from the center of earth will be

(a) $3R$ (b) $4R$ (c) $5R$ (d) $8R$

5. In the given arrangement a source is placed at the fixed end of the spring with spring constant 100 N/m and the receiver is attached to the moving end of the spring, which inturn is connected to a box of mass 4kg, which is oscillating horizontally with amplitude of oscillation 2*m*. If the source emits a frequency of 990 Hz then (speed of sound = 330m/s)



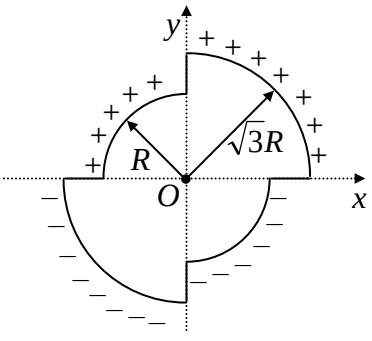
(a) maximum frequency received by the receiver is 990Hz

(b) minimum frequency received by the receiver is 990Hz

(c) frequency band width received by the receiver is 960Hz – 1020Hz

(d) maximum and minimum frequencies are received at the extreme positions.

6. In the figure there are four arcs carrying positive or negative charge all of them have charge density λ, then correct alternative is



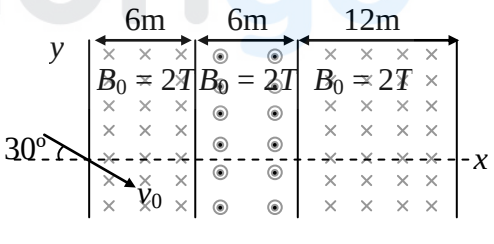
(a) the net dipole moment for the given charge distribution is $4\sqrt{5}\lambda R^2$

(b) the resultant electric field at the center and at any point on the axis passing through point *O* and perpendicular to the plane of arcs are opposite in direction

(c) if a uniform electric field is switched on perpendicular to the plane then the charge distribution starts rotating about x-axis

(d) potential at the center of the given charge distribution is non zero.

7. A particle having charge 1C and mass 1kg enters a region having uniform magnetic field of strength '2*T*' with a speed of 12 m/s, as shown in figure, then the correct statement is



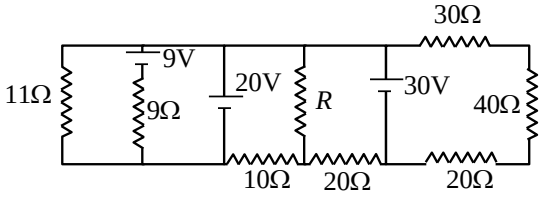
(a) the time for which the charge particle remains in magnetic field is 8.34s

(b) the velocity of charged particle becomes parallel to x-axis 6 times during its motion

(c) the shortest distance between the point where the charge particle enters the uniform magnetic field and the point where it emerges out is 6m

(d) the deviation of the charge particle when it emerges out of the magnetic field is '$2\pi/3$' rad.

8. In the given figure, the value of R so that power generated in R will be maximum is



(a) $\frac{45}{4}\Omega$ (b) $20\ \Omega$

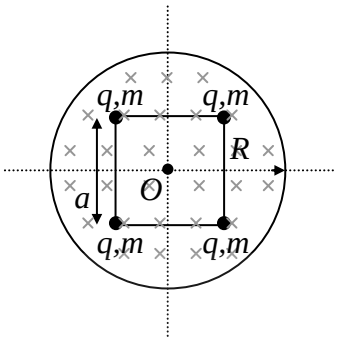
(c) $10\ \Omega$ (d) $\frac{20}{3}\Omega$

SECTION – II
Multiple Correct Choice Type

This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

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9. Four identical charge particles each of mass 0.1kg and charge $2C$ connected to each other via massless non-conducting rods of equal length. The whole arrangement is placed in a cylindrical region carrying a uniform magnetic field as shown in the figure ($B_0 = 4T, a = 1m$). Suddenly the magnetic field is switched off. Then choose the correct statement(s)



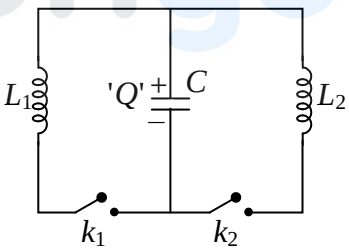
(a) the angular momentum of the system is 8 Nm/s

(b) the angular momentum of the system is $8\sqrt{2}$ Nm/s

(c) Angular velocity of the system is 40 rad/s

(d) the magnetic field at point O after the magnetic field is switched off is non zero.

10. The given arrangement carries a capacitor with capacitance 40mF and two inductors $L_1 = 25H$ and $L_2 = 100H$. If the capacitor initially carries a charge of 10 mC, then



(a) the maximum current through the inductor L_1 when key k_1 is closed is 20 mA

(b) the maximum current through the inductor L_2 when key k_2 is closed is 5mA

(c) the maximum current through inductor L_2 when both the keys are closed is $\sqrt{5}$ mA

(d) time period of oscillation of charge is minimum when both the keys are closed

11. Two identical point particles *A* and *B* are placed in front of a concave mirror of focal length 20cm, at distances 10 cm and 30 cm respectively. The particles oscillate perpendicular to the principal axis, such that the displacement equation for both the particles is given by $Y_A = Y_B = 0.1\sin(\pi t)$. The correct statement(s) is/are

(a) displacement equation of the image of particle *A* is $Y_A' = 0.2\sin(\pi t + \pi/2)$

(b) displacement equation of the image of particle *B* is $Y_B' = 0.3\sin(\pi t + \pi)$

(c) time period of oscillation of both the images is 2s.

(d) relative vertical displacement of image of A with respect to image of B is $Y_{AB}' = 0.4\sin(\pi t)$

12. With reference to the Young's double slit experiment, the correct statement(s) is/are

(a) if the separation between the screen and slits is increases, then the fringes inflate

(b) if the separation between the two slits in increased, then number of fringes on the screen decreases

(c) if the whole arrangement is immersed in a medium with refractive index μ, then the number of fringes on the screen will increase

(d) if the slits are illuminated with white light then the central maxima of the fringe pattern will also be white in colour

SECTION–III
Comprehension Type

This section contains 1 group of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

If two deuterium nuclei get close enough together, the attraction of the strong nuclear force will fuse them to make as isotope of helium. The process will release a vast amount of energy. The range of nuclear force is 10^{-15} m. This is the principle behind the nuclear fusion reactor. The deuterium nuclei moves so fast hence it is not possible to contain them by physical walls. Therefore, they are confined magnetically.

13. How fast two nuclei shall come to have a head on collision to fuse?

(a) 1.2×10^6 ms^{-1} (b) 2.6×10^6 ms^{-1} (c) 3.8×10^6 ms^{-1} (d) 8.3×10^6 ms^{-1}

14. The strength of magnetic field required to make deuterium nuclei moving with this speed, to be confined in a circle of diameter 2.5m

(a) 134 mT (b) 129 mT (c) 139 mT (d) 116 mT

15. The following is the process of formation of the Helium nuclei from the nuclear fusion of deuterium nuclei

$${}_1^2H + {}_1^2H \rightarrow {}_1^3H + {}_1^1H + \Delta E_1$$

$${}_1^3H + {}_1^2H \rightarrow {}_2^4He + {}_0^1n + \Delta E_2$$

[$m({}_1^2H) = 2.014102$ a.m.u, $m({}_1^3H) = 3.016049$ a.m.u, $m({}_2^4He) = 4.002603$ a.m.u, $m({}_0^1n) = 1.008665$ a.m.u, $m({}_1^1H) = 1.007825$ a.m.u] then $\Delta E_1 + \Delta E_2$ is

(a) 12.3 MeV (b) 21.6 MeV (c) 17.7 MeV (d) 25.6 MeV

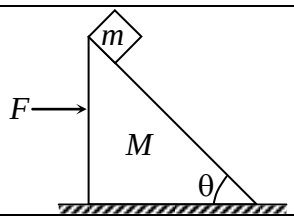
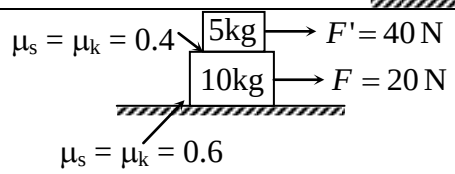
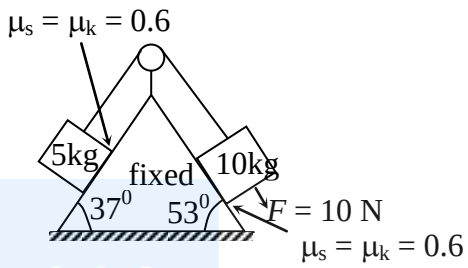
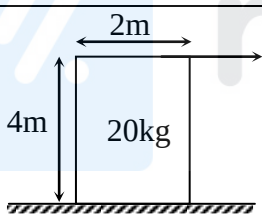
SECTION–IV
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

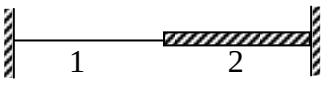
If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.

	1	2	3	4	5
A	●	○	○	●	●
B	○	●	●	○	○
C	●	●	○	○	○
D	○	○	○	●	●

1. In column I some arrangements of the objects are given. F represents the external constant force. In column II the possible variation of physical quantities are given. Match column I with column II ($\sin 37^0 = \frac{3}{5}$, $\cos 37^0 = \frac{4}{5}$)

Column I	**Column II**
(A) All surfaces are frictionless and $F = mg\sin\theta\cos\theta$ 	**1.** The centre of mass of the system accelerates in horizontal direction.
(B) 	**2.** The centre of mass of the system accelerates in vertical direction
(C) 	**3.** The object on which F is applied remains in translatory equilibrium
(D)  The box has uniform mass distribution	**4.** Kinetic energy of the system increases **5.** The friction force acting at all contact surfaces is equal to the corresponding limiting value of static friction.

2. Two strings 1 and 2 joined end to end and are fastened between two rigid supports under same tension. Mass per unit length of the strings are μ_1 and μ_2 respectively. A wave given as $y = A\sin(kx \pm \omega t)$ is generated in one of the string. A_r and A_t represents respectively the amplitudes of reflected and transmitted wave respectively from the interface. Match column I with column II.

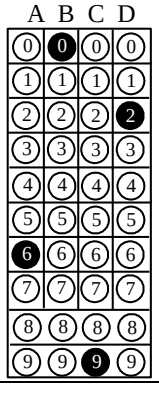


Column I	Column II
(A) $\mu_1 = 4\mu_2$ and incident wave is generated in string 1.	**1.** $A_r = \frac{A}{3}$ or $A_r = A$
(B) $\mu_1 = 4\mu_2$ and incident wave is generated in string 2.	**2.** $A_t = \frac{2A}{3}$ or $A_t = \frac{4A}{3}$
(C) $\mu_1 = \mu_2$ and incident wave is generated in string 1.	**3.** In either string, there will be a standing wave and a component of traveling wave.
(D) $\mu_1 << \mu_2$ and incident wave is generated in string 1.	**4.** In either string, there will be a standing wave. **5.** The energy of the system remains constant.

mathongo

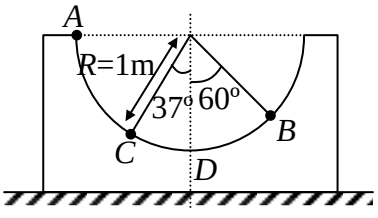
SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



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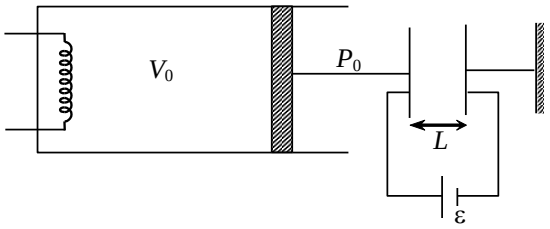
1. In the given arrangement there is a wedge of mass 10kg and a particle of mass 2kg, the friction co-efficient between the wedge and particle is 0.5, and there is no friction between the ground and wedge. When the particle is released from point *A*, it reaches point *B* on the wedge and comes rest momentarily, then it starts moving from point *B* and comes to rest at point *C*, momentarily if, then it starts from point *C* and finally comes to rest at point *D*. Calculate the magnitude of work done by friction force when it moves from point *B* and comes to momentary rest at point *C*?



Space for work

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2. An ideal gas is enclosed in a cylinder fitted with a frictionless piston. The piston is connected with a light rod to one plate of capacitor whose other plate is fixed as shown. Initially volume of the gas inside the cylinder is V_0. Atmospheric pressure is P_0, separation between the plates is L, area of the piston as well as of the capacitor plates is A and emf of battery is ε. A heater supplies heat to the gas so that pressure of the gas is given as $P = P_0 - \frac{n\varepsilon_0\varepsilon^2}{L^2}$, when piston is displaced by a distance $\frac{L}{2}$. Find value of n.



Space for work

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3. A man in a boat wishes to cross a 400 m wide river flowing at the rate of 2 m/s. Speed of boat in still water is 4 m/s. The person starts to row the boat perpendicular to the river flow from one bank of the river. When he reaches in the middle of the river. In order to reach the point directly opposite to the starting point on the opposite bank, he starts to row the boat at an angle θ with the vertical. The value of tan θ is $\frac{4}{n}$. Find value of *n*.

Space for work

mathongo

4. A tank has a hole area of cross-section *A*. Water from a pipe of inner cross-section 2*A* is entering the tank with a velocity of 5 m/s. calculate the height upto which the water will be filled inside the tank at steady state?

Space for work

mathongo

CHEMISTRY

SECTION – I
Single Correct Choice Type

This section contains 6 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

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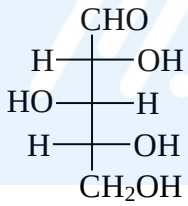
1. A 150 mL of solution of I_2 is divided into two unequal parts. 1st part reacts with hypo solution in acidic medium. 15 mL of 0.4 M hypo was consumed. 2nd part was added to 100 mL of 0.3 M NaOH solution. Residual base required 10 mL of 0.3 M H_2SO_4 solution for complete neutralization. What was the initial concentration of I_2?

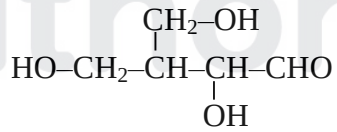
(a) 0.08 M (b) 0.1 M (c) 0.2 M (d) none of these

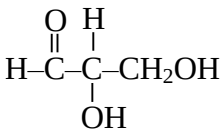
2. Metal 'M' + air $\xrightarrow{\Delta}$ (A) $\xrightarrow{H_2O}$ (B) $\xrightarrow{HCl}$ White fumes. Metal 'M' can be

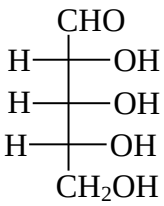
(a) Li, Mg or Al (b) Li, Al or K (c) Na, K or Mg (d) Li, Na or K

3. Compound (A) of molecular formula, $C_5H_{10}O_4$ is oxidised by Br_2/H_2O to, give acid (B) of molecular formula $C_5H_{10}O_5$ which readily forms a lactone on heating. (A) forms a triacetate with Ac_2O and an osazone with $PhNHNH_2$, (A) is oxidised by HIO_4. Only one mole of HIO_4 is consumed per mol of (A). The structure of (A) will be

(a)


(b)


(c)


(d)


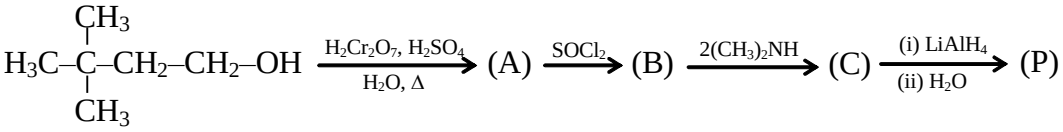
4. Consider the following reaction in aqueous solution at equilibrium.

$6HCHO(aq) \rightleftharpoons C_6H_{12}O_6(aq)$

What is the degree of association of HCHO in the above reaction if observed molar mass of HCHO and $C_6H_{12}O_6$ in the mixture is 150.

(a) 0.5 (b) 0.833 (c) 0.9 (d) 0.96

5. Identify product (P) in the following reaction sequence.



(a) $(CH_3)_3C–CH_2CN$

(b) $(CH_3)_3C–CH_2–CH_2N(CH_3)_2$

(c) $(CH_3)_3C–CH_2–\underset{\displaystyle N(CH_3)_2}{\underset{|}{CH}}–N(CH_3)_2$

(d) $(CH_3)_3C–CH_2–\underset{\displaystyle OH}{\underset{|}{CH}}–N(CH_3)_2$

6. Match List-I with List-II and select the correct answer

List-I	**List-II**
(A) $AlCl_3$ if $\propto = 0.8$	**1.** $i = 3.4$
(B) $BaCl_2$ if $\propto = 0.9$	**2.** $i = 2.8$
(C) Na_3PO_4 if $\propto = 0.9$	**3.** $i = 3.8$
(D) $K_4[Fe(CN)_6$ if $\propto = 0.7$	**4.** $i = 3.7$

	(A)	**(B)**	**(C)**	**(D)**
(a)	1	2	4	3
(b)	2	1	3	4
(c)	1	4	2	3
(d)	4	2	3	1

SECTION – II
Multiple Correct Choice Type

This section contains 5 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

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7. The radial distribution function [P(r)] is used to determine the most probable radius which is used to find the electron in a given orbital. $\frac{dP(r)}{dr}$ for 1s-obtital of hydrogen like atom having atomic number 'Z' is

$$\frac{dP(r)}{dr} = \frac{4Z^3}{a_o^3}\left[2r - \frac{2Zr^2}{a_o}\right]e^{-2Zr/a_o}$$

then which of the following statements is/are correct?

(a) At the point of maximum value of radial distribution function $\frac{dP(r)}{dr}$ = zero, one antinode is present.

(b) Most probable radius of Li^{2+} is $\frac{a_o}{3}$ pm.

(c) Most probable radius of He^+ is $\frac{a_o}{2}$ pm.

(d) Most probable radius of hydrogen ion (H^+) is a_o pm.

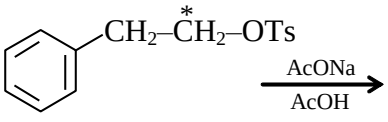
8. The possible product(s) formed in the following reaction is/are

$Cl_2O_6 + NaOH \longrightarrow ?$

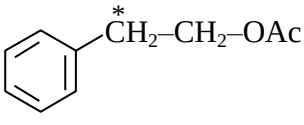
(a) $NaClO_4$ (b) $NaOCl$

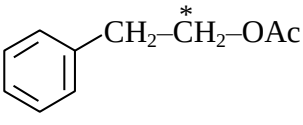
(c) $NaClO_2$ (d) $NaClO_3$

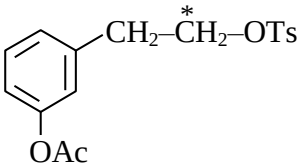
9. Product(s) formed during this reaction are

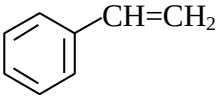


[C* = isotopic carbon]

(a) 

(b) 

(c) 

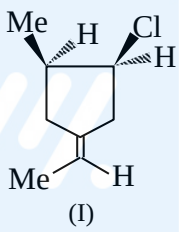
(d) 

10. In which of the following pair(s), the minerals are converted into metals by self reduction?

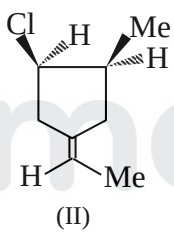
(a) Ag_2S, ZnS (b) Cu_2S, HgS

(c) PbS, ZnS (d) Cu_2S, PbS

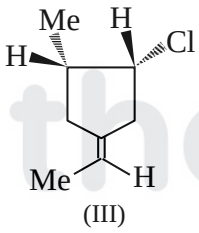
11. Consider the structures shown below:



(I)



(II)



(III)

and select the correct choice(s).

(a) (I) and (II) are geometrical isomer. (b) (I) and (III) are enantiomers.

(c) (II) and (III) are homomers. (d) (I) and (II) are enantiomers.

SECTION – III

Reasoning Type

This section contains 5 reasoning type questions. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True; **Statement-2** is a correct explanation for **Statement-1**.

(B) **Statement-1** is True, **Statement-2** is True; **Statement-2** is not a correct explanation for **Statement-1**.

(C) **Statement-1** is True, **Statement-2** is False.

(D) **Statement-1** is False, **Statement-2** is True.

12. **Statement-1:** An optically active molecule must be asymmetric.

Statement-2: Asymmetric molecules will have non-superposable mirror image.

(a) A (b) B (c) C (d) D

13. **Statement-1:** For a real gas 'X', the Joule-Thomson coefficient has non-zero value.

Statement-2: The gas 'X' has intermolecular interactions.

(a) A (b) B (c) C (d) D

14. **Statement-1:** A certain reaction with $\Delta_r G° > 0$ will be non-spontaneous.

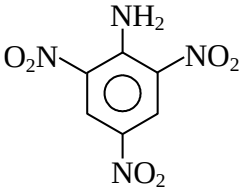
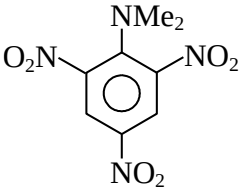
Statement-2: $\Delta_r G°$ can be used to predict the spontaneity of any reaction under standard conditions.

(a) A (b) B (c) C (d) D

15. **Statement-1:** The number of nodes in the radial function is equal to n-*l*-1.

Statement-2: The point at which r = 0, is not considered as node.

(a) A (b) B (c) C (d) D

16. **Statement-1:**  is less basic than  .

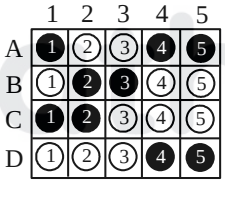
Statement-2: Only the +I effect of two methyl groups on nitrogen makes N,N-dimethylpicramide more basic than picramide.

(a) A (b) B (c) C (d) D

SECTION–V
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.

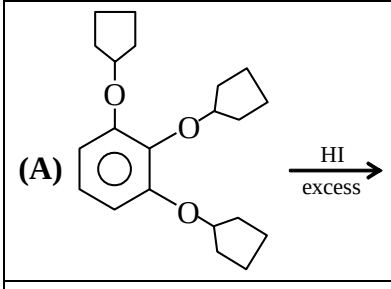
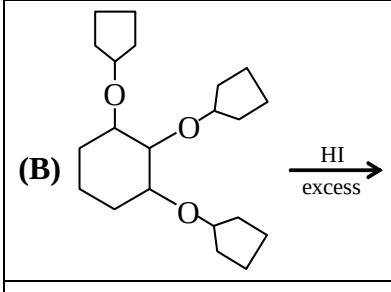
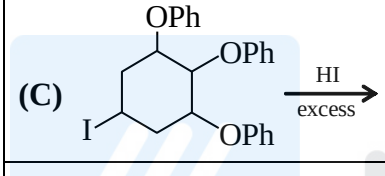
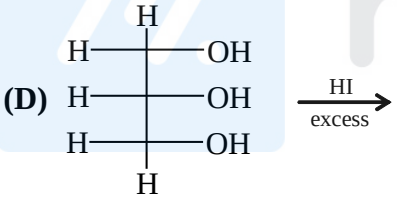
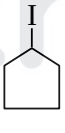


1.

Column I	Column II
(A) Reversible cooling of an ideal gas at constant volume.	**1.** $w = 0, q < 0, \Delta U < 0$
(B) Reversible isothermal expansion of an ideal gas.	**2.** $w = 0, q > 0, \Delta U > 0$
(C) Adiabatic expansion of non-ideal gas into vacuum.	**3.** $w = 0, q = 0, \Delta U = 0$
(D) Reversible melting of sulphur at normal melting point.	**4.** $w < 0, q > 0, \Delta U = 0$ **5.** $\Delta H \neq 0$

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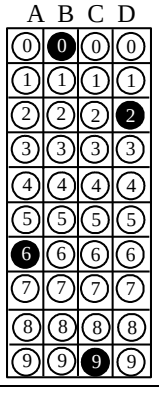
2.

Column I	Column II
(A) 	**1.** More than 4-moles of HI will be consumed.
(B) 	**2.** Product will react with Na.
(C) 	**3.** Product formed contains 2° halides.
(D) 	**4.**  is one of the product.
	5. One of the products contains more than one I-atoms.

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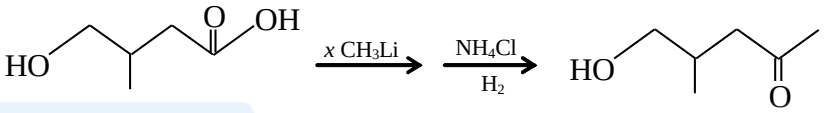
SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



---

1. When carboxylic acid reacts with organolithium reagents to give ketones, side reactions sometime occurs, for example.



value of (x) in the above reaction is

Space for work

mathongo

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2. If the slope of 'Z' (compressibility factor) v/s 'P' curve is constant $\left(\text{slope} = \frac{\pi}{492.6}\text{atm}^{-1}\right)$ at a particular temperature (300 K) and very high pressure, then calculate diameter of the molecule (in Å) ($N_A = 6 \times 10^{23}$, R = .0821 atm L, $\text{mol}^{-1}\ \text{K}^{-1}$)

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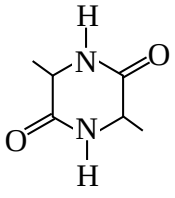
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3. A fresh radioactive mixture containing short lived species X and Y. Both the species together emitting 8000 α-particles per minute initially. 20 minutes later X was emitting α-particles at the rate of 4500 per minute. If the half-lives of the species X and Y are 10 minute and 500 hours, then what is the ratio of initial activities of X and Y in the mixture?

Space for work

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4. The number of stereoisomer formed by the given compound is



Space for work

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MATHEMATICS

SECTION – I
Single Correct Choice Type

This section contains 8 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

---

1. In a parallelogram $ABCD$, $\left|\overrightarrow{AB}\right| = a$, $\left|\overrightarrow{AD}\right| = b$ and $\left|\overrightarrow{AC}\right| = c$, then $\overrightarrow{DB}.\overrightarrow{AB}$ has value

(a) $\frac{3a^2 + b^2 - c^2}{2}$ (b) $\frac{a^2 + 3b^2 - c^2}{2}$

(c) $\frac{a^2 - b^2 + 3c^2}{2}$ (d) $\frac{a^2 + 3b^2 + c^2}{2}$

2. If $A > 0$, c, d, u, v are non zero constants, and the graphs of $f(x) = |Ax + c| + d$ and $g(x) = -|Ax + u| + v$ intersect exactly at two points (1, 4) and (3, 1), then the value of $\frac{u + c}{A}$ equals

(a) 4 (b) -4 (c) 2 (d) -2

3. If the mapping $f(x) = ax + b$, $a > 0$ maps $[-1, 1]$ onto $[0, 2]$, then $\cot(\cot^{-1}7 + \cot^{-1}8 + \cot^{-1}18)$ is

(a) $f(-1)$ (b) $f(0)$ (c) $f(1)$ (d) $f(2)$

4. If $I_1 = \int_1^{\sin\theta} \frac{x}{1+x^2}dx$ and $I_2 = \int_1^{\operatorname{cosec}\theta} \frac{dx}{x(x^2+1)}$, then the value of $\begin{vmatrix} I_1 & I_1^2 & I_2 \\ e^{I_1+I_2} & I_2^2 & -1 \\ 1 & I_1^2 + I_2^2 & -1 \end{vmatrix}$ is

(a) 1 (b) $\frac{-11}{2}$ (c) 9 (d) 0

5. If $[\sin x] + [\sqrt{2}\cos x] = -3$, $x \in [0, 2\pi]$ ($[\cdot]$ denotes the greatest integer function), then x belongs to

(a) $\left(\pi, \frac{5\pi}{4}\right)$ (b) $\left[\frac{5\pi}{4}, 2\pi\right]$ (c) $\left(\frac{5\pi}{4}, 2\pi\right)$ (d) none of these

6. The value of $\lim\limits_{x\to\infty} \frac{\cot^{-1}(x^{-a}\log_a x)}{\sec^{-1}(a^x \log_x a)} (a>1)$ is equal to

(a) 1 (b) 0 (c) $\frac{\pi}{2}$ (d) does not exist

7. Tangents PT_1 and PT_2 are drawn from a point P to the circle $x^2 + y^2 = a^2$. If the point P lies on the line $px + qy + r = 0$, then the locus of the centre of circumcircle of the triangle PT_1T_2 is

(a) $px + qy = r$ (b) $(x-p)^2 + (y-q)^2 = r^2$

(c) $px + qy = \frac{r}{2}$ (d) $2px + 2py + r = 0$

8. Let $f(x) = \lim\limits_{n\to\infty} \sum\limits_{r=0}^{n-1} \frac{x}{(rx+1)\{(r+1)x+1\}}$, then

(a) $f(x)$ is continuous but not differentiable at $x = 0$

(b) $f(x)$ is both continuous and differentiable at $x = 0$

(c) $f(x)$ is neither continuous nor differentiable at $x = 0$

(d) $f(x)$ is a periodic function

SECTION – II
Multiple Correct Choice Type

This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

---

9. If $A(z_1)$, $B(z_2)$ and $C(z_3)$ are three points in argand plane where $|z_1+z_2|=|z_1|-|z_2|$ and $|(1-i)z_1+iz_3|=|z_1|+|z_3-z_1|$, then

(a) *A*, *B* and *C* lie on a fixed circle with centre $\left(\frac{z_2+z_3}{2}\right)$

(b) *A*, *B*, *C* form right angle triangle

(c) *A*, *B*, *C* form an equilateral triangle

(d) *A*, *B*, *C* form an obtuse angle triangle

10. If the sides of a triangle vary slightly in such a way that its circumradius remains constant, then $\frac{da}{\cos A}+\frac{db}{\cos B}+\frac{dc}{\cos C}$ is (where *A*, *B*, *C* are angles of a ΔABC, and *a*, *b*, *c* are sides opposite to them, *R* is circumradius)

(a) $6R$ (b) $2R$

(c) 0 (d) $2R\,(dA+dB+dC)$

11. If $f(x)=\lim\limits_{n\to\infty} e^{x\tan\left(\frac{1}{n}\right)\ln\left(\frac{1}{n}\right)}$ and $\int\frac{f(x)dx}{\sqrt[3]{\sin^{11}x\cos x}}=g(x)+C$, then

(a) $g\left(\frac{\pi}{4}\right)=\frac{3}{2}$ (b) $g(x)$ is continuous for all x

(c) $g\left(\frac{\pi}{4}\right)=\frac{-15}{8}$

(d) $g(x)$ is non differentiable at infinitely many points

12. If *A*, *B*, *C* are three points with position vectors $\hat{i}+\hat{j}, \hat{i}-\hat{j}$ and $p\hat{i}+q\hat{j}+r\hat{k}$ respectively, then the points are collinear if

(a) $p=q=r=1$ (b) $p=q=r=0$

(c) $p=q, r=0$ (d) $p=1, q=2, r=0$

SECTION–III
Comprehension Type

This section contains 1 group of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

A parabola touches co-ordinate axis at $\left(\frac{25}{3}, 0\right)$ and $\left(0, \frac{25}{4}\right)$, then

13. Focus of parabola is

(a) (4, 3) (b) (2, 1) (c) (3, 4) (d) (1, 2)

14. Length of latus rectum is

(a) $\frac{41}{5}$ (b) $\frac{44}{5}$ (c) $\frac{48}{5}$ (d) $\frac{49}{5}$

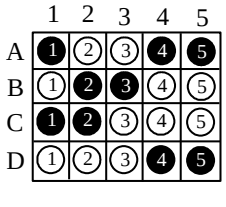
15. Equation of directrix is

(a) $4x+3y=0$ (b) $2x+y=0$ (c) $3x+4y=0$ (d) $x+2y=0$

SECTION–IV
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.



1.

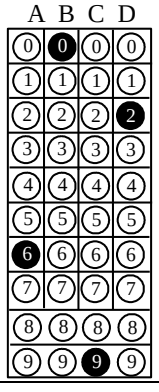
Column- I	Column -II
(A) $\lim\limits_{x\to\frac{\pi}{2}}\left[\frac{\lvert\sin x\rvert-\lvert\cos x\rvert+1}{3}\right]$ lies in (where [.] represents greatest integer function)	**1.** $(-2, 1)$
(B) f is a function satisfying $f(x+y)=f(x).f(y)$ and $f'(0)=2$. If $f\left(\frac{x}{2}\right)=kx^2$ has exactly one solution, then k is subset of	**2.** $(0, 3)$
(C) $\int_{-2}^{2}\max([x], [-x])dx$ lies in the interval (where [.] represents greatest integer function)	**3.** $(1, 3)$
(D) Let $f(x)=\cos^{-1}(\cos x)$ and $g(x)=\sin^{-1}[x+1]+\cos^{-1}[x]$ (where [.] represents greatest integer function), then the number of solution of $f(x)+g(x)=3$ lies in the interval	**4.** $(-2, 3)$ **5.** $(1, 2)$

2.

Column- I	Column -II
(A) Given that, $(1+\tan1°)(1+\tan2°)\ldots\ldots(1+\tan45°)=2^{\lambda}$, then $(\lambda+1)$ is divisible by	**1.** –2
(B) If the system of equations $x+2y+3z=7$, $2x-y+z=5$ and $x-3y-\lambda z=1$ has no solution, then λ may be	**2.** 0
(C) The value of λ for which $\frac{x-1}{2}=\frac{y+\lvert\lambda\rvert}{3}=\frac{z}{1}$ lies in the plane $2x-y-z=5$ is/ are	**3.** 3
(D) Plane $ax+y+z=a$ intersects x-axis at A, y-axis at B and z-axis at C and area of ΔABC is $\sqrt{6}$ unit, then 'a' may be	**4.** 2 **5.** –3

SECTION–V
Integer Answer Type

This section contains 4 questions. The answer to each of the questions is a **single digit integer,** ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers A, B, C and D (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the following:



---

1. If $2f(x)+f(-x)=\frac{1}{x}\sin\left(\frac{x^2-1}{x}\right)$ find $\int_{1/e}^{e} f(x)\,dx$.

Space for work

mathongo

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2. The probability of a man hitting a target in one fire is $1/5$. Find the minimum number of fire; he must follow in order to make his chances of hitting the target more than $\frac{3}{4}$.

Space for work

mathongo

3. Consider two polynomials $f(x)$ and $g(x)$ as $g(x)=\sum_{r=0}^{200}\alpha_r x^r$ and $f(x)=\sum_{r=0}^{200}\beta_r x^r$.

Given (i) $\beta_r=1\,\forall r\geq 100$ (ii) $f(x+1)=g(x)$. Let $A=\sum_{r=100}^{200}\alpha_r$. Find the remainder when A is divided by 15.

Space for work

mathongo

4. The integers from 1 to 1000 are written in order around a circle. Starting at 1, every fifteenth number is marked (that is 1, 16, 31 etc.). This process is continued until a number is reached which has already been marked. If the number of marked numbers is A, then find $\frac{A}{50}$.

Space for work

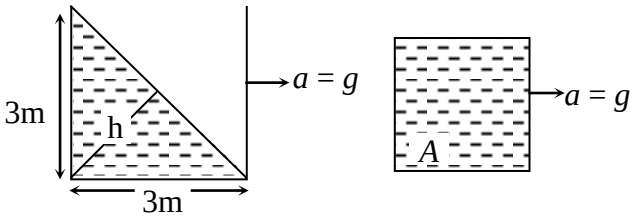
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PAPER-I
PHYSICS
SECTION – I
Single Correct Choice Type

1. $P_1 = \rho\sqrt{g^2 + a^2}\,h = \rho\sqrt{2g}\,\frac{3}{\sqrt{2}} = 3\rho g$

$P_2 = \rho g h' + \rho a l = 3\rho g + 3\rho g = 6\rho g$

$\frac{P_1}{P_2} = \frac{1}{2}$

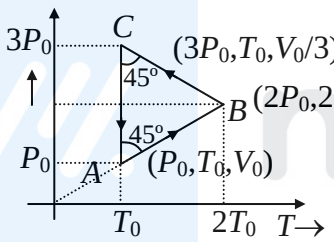


$\therefore$ **(d)**

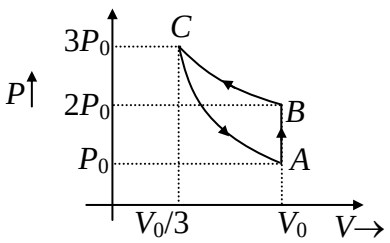
2. $1200 \times (g + a) = \frac{1}{3} \times 2.4 \times 10^8 \times 3 \times 10^{-4}$ or $a = 10\,\text{ms}^{-2}$

$\therefore$ **(b)**

3. AB is an isochoric process and CA is an isothermal process



Corresponding P-V curve for the above cyclic process.



$\therefore$ **(b)**

4. According to the principle of conservation of angular momentum

$m\sqrt{\frac{10GM}{6R}}R = mv'(R + h)$ … (1)

Applying principle of conservation of mechanical energy

$-\frac{GMm}{R} + \frac{1}{2}m\left(\frac{10GM}{6R}\right) = -\frac{GMm}{R+h} + \frac{1}{2}mv'^2$ … (2)

Solving the above equations we get $h = 4R$

$\therefore$ **(c)**

5. The maximum and minimum frequency is received by the receiver when it is at the mean position.

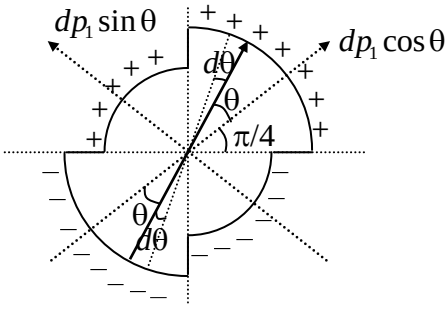
$$f_{max} = \left(\frac{v + v_0}{v}\right) f_0 = 1020\,\text{Hz}$$

$$f_{min} = \left(\frac{v - v_0}{v}\right) f_0 = 960\ \text{Hz}$$

Frequency band width received by the receives ranges between 960 Hz – 1020 Hz.

$\therefore$ **(c)**

6. Dipole moment of the two bigger arcs is



$$\int dP_1 = \int_{-\pi/4}^{\pi/4} dq(2\sqrt{3}R)\cos\theta$$

$$\int dP_1 = \int_{-\pi/4}^{\pi/4} 6\lambda R^2 \cos d\theta$$

$$P_1 = 6\lambda R^2 \int_{-\pi/4}^{\pi/4} \cos\theta d\theta$$

$$= 6\sqrt{2}\lambda R^2$$

Similarly for smaller arcs the dipole moment is

$$P_2 = 2\sqrt{2}\ \lambda R^2$$

$$|\vec{P}_{net}| = \sqrt{P_1^2 + P_2^2} = 4\sqrt{5}\lambda R^2$$

$\therefore$ **(a)**

7. Time during which the charged particle remains in magnetic field is

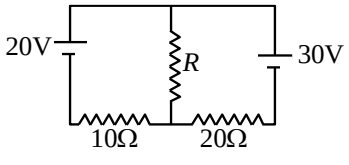
$$T = \frac{8\pi m}{3qB} = 4.15s$$

$$\delta = \frac{4\pi}{3} \Rightarrow \text{deviation of the charged particle}$$

Shortest distance between the entry and exit point is $= \sqrt{3}R = 6\sqrt{3}\,\text{m}$

$\therefore$ **(b)**

8. The given figure can be redrawn is



For P_{max}, $R = \frac{10 \times 20}{10 + 20}\Omega = \frac{20}{3}\Omega$

$\therefore$ **(d)**

SECTION – II
Multiple Correct Choice Type

9. According to Newton's second law of motion

$$\tau = \frac{dL}{dt} \text{ or } \int_0^t \tau dt = L_f - L_i$$

$$\Rightarrow \quad \int 4qE\frac{a}{\sqrt{2}}dt = L_f$$

$$qa^2B_0 = 2ma^2\omega$$

$$\omega = \frac{qB_0}{2m}$$

$\therefore$ **(a, c, d)**

10. Maximum current through the inductor is $I_{\max} = \frac{Q_0}{\sqrt{LC}}$

When only switch k_1 is closed then $I_{\max} = \frac{10\times10^{-3}}{5\times2\times10^{-1}} = 10mA$

When only switch k_2 is closed then $I_{\max} = \frac{10\times10^{-3}}{10\times2\times10^{-1}} = 5mA$

When both the switches are closed then maximum current through L_2 is

$$I_2 = Q_0\sqrt{\frac{L_1}{L_2(L_1+L_2)C}} = \sqrt{5}mA$$

$\therefore$ **(b, c, d)**

11. Displacement of image of particle *A* is in phase with the displacement of particle *A* hence the displacement equation of particle *A*'s image is

$$Y_A' = 0.2\sin(\pi t) \qquad \ldots (1)$$

Displacement of image of particle *B* is out of phase with the displacement of particle *B* therefore the displacement of equation of particle *B*'s image is

$$Y_B' = 0.2\sin(\pi t + \pi) \qquad \ldots (2)$$

Relative vertical displacement of *A* w.r.t. to *B* is

$$Y_{AB}' = Y_A' - Y_B'$$

$$= 0.2\sin(\pi t) - 0.2\sin(\pi t + \pi)$$

$$= 0.4\sin(\pi t) \qquad \ldots (3)$$

$\therefore$ **(c, d)**

12. Fringe width $= \dfrac{\lambda D}{d}$

If D increases then fringe width increases and fringes inflate λ decreases when the whole system is immersed in a liquid of refractive index μ then fringe width decreases fringes increases.

$\therefore$ **(a, c, d)**

SECTION–III
Comprehension Type

13. $r = \dfrac{q_1 q_2}{4\pi\varepsilon_0 (2KE)}$

$v^2 = \dfrac{q_1 q_2}{4\pi\varepsilon_0 mr}$

$v^2 = \dfrac{1.6\times10^{-19}\times1.6\times10^{-19}\times9\times10^{9}}{10^{-15}\times2\times1.67\times10^{-27}}$

$v = 8.3 \times 10^6$ ms^{-1}

$\therefore$ **(d)**

14. $R = \dfrac{mv}{eB}$

or $B = \dfrac{2\times1.67\times10^{-27}\times8.3\times10^{6}}{1.6\times10^{-19}\times1.25} = 139mT$

$\therefore$ **(c)**

15. $\Delta E_1 + \Delta E_2 = \dfrac{(0.023213)\times1.66\times10^{-27}\times(3\times10^{8})^2}{1.6\times10^{-19}} = 21.6MeV$

$\therefore$ **(b)**

SECTION–IV
Matrix–Match Type

1. **(A) – (1, 2, 3, 4); (B) – (1, 3, 4); (C) – (3, 5); (D) – (1, 2, 3, 4)**

2. **(A) – (1, 2, 3, 5); (B) – (1, 2, 3, 5); (C) – (1, 4, 5); (D) – (1, 4, 5)**

SECTION–V
Integer Answer Type

1. Work done by friction is simply the changes in total energy of the system as the particle moves from point B to point C

$W_f = T.E._i - T.E_f = MgR\cos37^\circ - (MgR\cos60^\circ)$

$W_f = 0.3MgR$

$W_f = 6\,\text{J}$

Ans.: 6

2. Let the piston be displaced by x**.**

$$PA + \frac{q^2}{2\varepsilon_0 A} = P_0 A$$

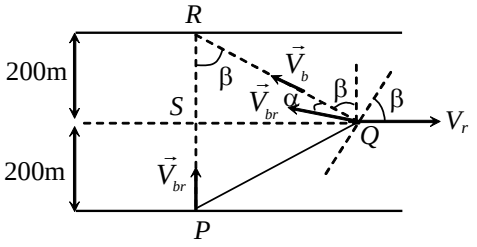
$$PA + \frac{1}{2\varepsilon_0 A}\frac{\varepsilon_0^2 A^2}{(L-x)^2}\varepsilon^2 = P_0 A$$

$$P = P_0 - \frac{\varepsilon_0 \varepsilon^2}{2(L-x)^2} = P_0 - \frac{2\varepsilon_0 \varepsilon^2}{L^2}$$

$\therefore \quad n = 2$

Ans.: 2

3. To reach point R from Q, the boat should be aimed at point R as seen from ground



$$\vec{V}_r \cos\beta = \vec{V}_{br} \sin\alpha$$

$$2 \times \frac{2}{\sqrt{5}} = 4\sin\alpha$$

$$\sin\alpha = \frac{1}{\sqrt{5}}$$

$$\Rightarrow \quad \tan\alpha = \frac{1}{2}$$

Also $\quad \tan\beta = \frac{1}{2}$

$$\theta = \alpha + \beta$$

$$\Rightarrow \quad \tan\theta = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta} = \frac{4}{3}$$

$\therefore \quad n = 3$

Ans.: 3

4. At steady state the rate at which the liquid is entering the tank should be equal to the rate at which the liquid is flowing out of the tank therefore

$$A\sqrt{2gh} = 2Av$$

$$h = \frac{2v^2}{g} = 5\,\text{m}$$

Ans.: 5

CHEMISTRY

SECTION – I

Single Correct Choice Type

1. **(b)** $I_2 + 2Na_2S_2O_3 \longrightarrow 2NaI + Na_2S_4O_6$

Millimoles of $Na_2S_2O_3$ consumed = 6.

Millimoles of I_2 consumed = 3.

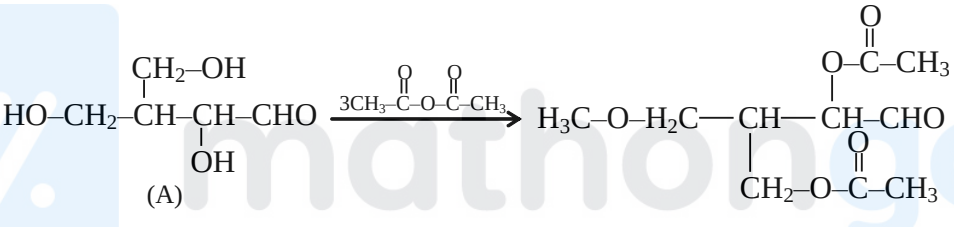
$3I_2 + 6NaOH \longrightarrow 5NaI + NaIO_3 + 3H_2O$

Millimole of I_2 reacted with NaOH = $\frac{30-6}{2} = 12$

Total millimole of I_2 consumed in this reaction = 15.

Initial concentration of I_2 = $\frac{15}{150}$ = **0.1 M**.

2. **(a)** Compound (A) is a metal nitride and (B) is NH_3 which gives white fumes of NH_4Cl with HCl. The metals which directly react with N_2 present in air on heating are Li, Mg and Al.

3. **(b)** 

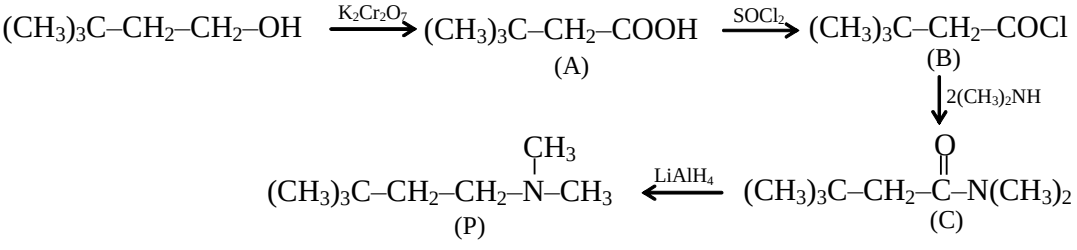
One mol of this compound can react with only 1 mol of HIO_4. It is oxidised by Br_2/H_2O to give carboxylic acid which forms a lactone ring of five member on heating.

4. **(d)** $6HCHO \rightleftharpoons C_6H_{12}O_6$

$c(1-\alpha)$ $\quad\quad$ $\frac{c\alpha}{6}$

$$\frac{M_{theoretical}}{M_{observed}} = \frac{30}{150} = \frac{c(1-\alpha)+\frac{c\alpha}{6}}{c}$$

α = **0.96**.

5. **(b)** 

6. **(a)** $AlCl_3 \rightleftharpoons Al^{3+} + 3Cl^-$

$$\underset{1-0.8}{} \quad \underset{0.8}{} \quad \underset{3\times 0.8}{}$$

$$i = \frac{(3\times 0.8) + 0.8 + (1-0.8)}{1} = 3.4$$

$$\underset{1-0.9}{BaCl_2} \rightleftharpoons \underset{0.9}{Ba^{2+}} + \underset{2\times 0.9}{2Cl^-}$$

Proceeding in the same way $i = 3.7$ for Na_3PO_4 and $i = 3.8$ for $K_4[Fe(CN)_6]$.

SECTION – II
Multiple Correct Choice Type

7. **(a, b, c)**

At the point of maximum value of radius distribution function $\frac{dp(r)}{dr} = 0$.

$$\therefore \quad 2r - \frac{2r^2 z}{a_o} = 0$$

$$r = \frac{a_o}{z} \text{ pm.}$$

Radius of $Li^{2+} = \frac{a_o}{3}$ pm.

Radius of $He^+ = \frac{a_o}{2}$ pm.

Radius of H^+ ion is the radius of nucleus not the radius of shell.

8. **(a, b)** $3Cl_2O_6 + 6NaOH \longrightarrow 5NaClO_4 + NaOCl + 3H_2O$

9. **(a, b)** Due to rearrangement (neighbouring group participation by phenyl ring).

10. **(b, d)** $Cu_2S + 2Cu_2O \longrightarrow 6Cu + SO_2$

$HgS + 2HgO \longrightarrow 3Hg + SO_2$

$PbS + 2PbO \longrightarrow 3Pb + SO_2$

11. **(b, c, d)**

Structure (II) is non-superposable mirror image of (I). Structure (II) can be obtained with similar configuration when the molecule (III) is rotated out of plane by 180°. Therefore (II) and (III) are identical.

SECTION – III
Reasoning Type

12. **(d)** An optically active molecule may either be asymmetric or dissymmetric.

13. **(a)** For ideal gas Joule Thomson coefficient has zero value. Intermolecular interaction can be of attractive or repulsive nature.

14. **(d)** Any reaction can be under standard or non-standard conditions. If the standard conditions are used, statement-1 is correct or if non-standard conditions are employed, it may be incorrect.

15. **(b)**

16. **(c)** Due to steric inhibition of resonance, the basicity of N-N-dimethylpicramide is more than picramide. +I effect is just a supporting factor but the main factor is steric factor.

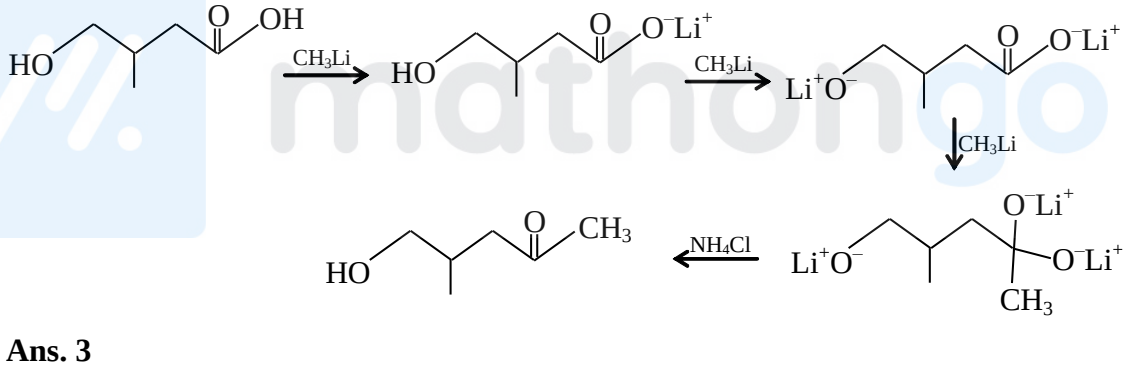
SECTION–IV
Matrix–Match Type

1. **(A) – 1, 5 ; (B) – 4 ; (C) – 3 ; (D) – 2, 5**

2. **(A) – 2, 4 ; (B) – 1, 2, 3, 4, 5 ; (C) – 2, 3, 5 ; (D) – 1, 2, 3**

SECTION–V
Subjective or Numerical Type

1.



Ans. 3

2. $Z = 1 + \frac{Pb}{RT}$ (at high pressure)

$$\text{Slope} = \frac{b}{RT} = \frac{\pi}{492.6}$$

$$b = \frac{\pi}{492.6} \times 0.0821 \times 300$$

$$\text{and } b = \frac{4}{3}\pi r^3 \times 4N_A$$

$$= \frac{\pi}{492.6} \times 0.0821 \times 300 \times 10^{-3}$$

$r = 2.5$ Å ; $d = 5$ Å

Ans. 5

3. Let initial activities of X and Y be x_o and y_o. After 2 half-lives the activity of x will be $\frac{x_o}{4}$

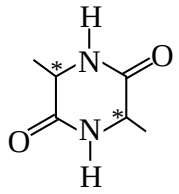
$x_o + y_o = 8000$; $\frac{3x_o}{4} = 4500$

$x_o = 6000$; $y_o = 2000$

$\frac{x_o}{y_o} = 3.$

Ans. 3

4.



This is symmetrical and has 2-chiral centres so

$2^{n-1} + 2^{\frac{n-2}{2}} = 3$ isomers.

Ans. 3

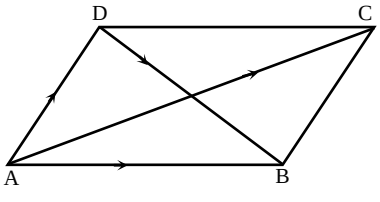
MATHEMATICS

SECTION – I

Single Correct Choice Type

1. (a) $\overrightarrow{DB} = \overrightarrow{DA} + \overrightarrow{AB}$ or $\overrightarrow{DA} = \overrightarrow{DB} - \overrightarrow{AB}$

$\therefore \quad \left(\overrightarrow{DA}\right)^2 = \left(\overrightarrow{DB}\right)^2 + \left(\overrightarrow{AB}\right)^2 - 2\overrightarrow{DB}.\overrightarrow{AB} \qquad \ldots(i)$



In parallelogram

$2\left(a^2 + b^2\right) = c^2 + DB^2 \Rightarrow \left(DB\right)^2 = 2a^2 + 2b^2 - c^2$

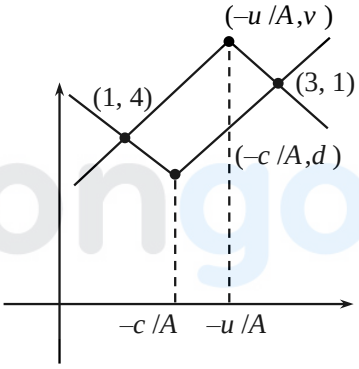
From (i) $\Rightarrow \quad b^2 = 2a^2 + 2b^2 - c^2 + a^2 - 2\overrightarrow{AB}.\overrightarrow{DB}$

$\overrightarrow{AB} \,.\, \overrightarrow{DB} = \dfrac{3a^2 + b^2 - c^2}{2}$

2. (b) The figure is a parallelogram and the diagonals bisect each other:

$\therefore \quad \left(\dfrac{-u}{A}\right) + \left(\dfrac{-c}{A}\right) = 3 + 1;$

$\therefore \quad \dfrac{u + c}{A} = -4$



3. (d) $f(x) = ax + b \qquad f'(x) = a > 0$

$\therefore \quad f(x)$ is an increasing function.

$\Rightarrow \quad f(-1) = 0$ and $f(1) = 2 \Rightarrow -a + b = 0;\ a + b = 2 \quad \Rightarrow \quad f(x) = x + 1$

Now $\cot\left(\cot^{-1}7 + \cot^{-1}8 + \cot^{-1}18\right)$

$= \cot\left(\tan^{-1}\dfrac{1}{7} + \tan^{-1}\dfrac{1}{8} + \tan^{-1}\dfrac{1}{18}\right) = \cot\left(\tan^{-1}\left(\dfrac{3}{11}\right) + \tan^{-1}\left(\dfrac{1}{18}\right)\right)$

$= \cot\left(\tan^{-1}\left(\dfrac{1}{3}\right)\right) = \cot\left(\cot^{-1}3\right) = 3 = f(2).$

4. (d) $I_2 = \displaystyle\int_1^{\operatorname{cosec}\theta} \frac{dx}{x\left(x^2+1\right)} = -\int_1^{\sin\theta} \frac{t}{1+t^2}dt$

$I_2 = -I_1$

$$\therefore \quad \begin{vmatrix} I_1 & I_1^2 & I_2 \\ e^{I_1+I_2} & I_2^2 & -1 \\ 1 & I_1^2+I_2^2 & -1 \end{vmatrix} = 0$$

5. **(a)** $[\sin x] = -1$ and $[\sqrt{2}\cos x] = -2$

or $-1 \le \sin x < 0$ and $-2 \le \sqrt{2}\cos x < -1$

$x \in (\pi, 2\pi)$ and $x \in \left(\frac{3\pi}{4}, \frac{5\pi}{4}\right) \Rightarrow \quad x \in \left(\pi, \frac{5\pi}{4}\right).$

6. **(a)** $I = \lim\limits_{x\to\infty} \dfrac{\cot^{-1}\left(\dfrac{\log_a x}{x^a}\right)}{\sec^{-1}\left(\dfrac{a^x}{\log_a x}\right)}$ as $\left(\dfrac{\log_a x}{x^a}\right) \to 0$ and $\left(\dfrac{a^x}{\log_a x}\right) \to \infty$

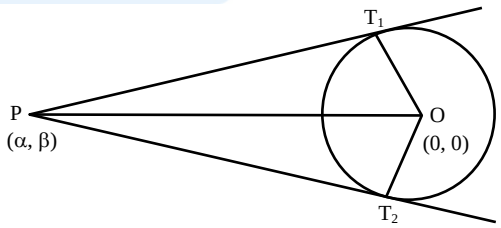
$\therefore \quad I = \dfrac{\pi}{2} \Big/ \dfrac{\pi}{2} = 1.$

7. **(d)** P, T_2, O, T_1 are concyclic points with PO as diameter

$\Rightarrow$ The circumcentre of ΔPT_1T_2 is $\left(\frac{\alpha}{2}, \frac{\beta}{2}\right)$

Now (α, β) lies on $px + qy + r = 0$

$\Rightarrow$ Locus of $\left(\frac{\alpha}{2}, \frac{\beta}{2}\right)$ is $2px + 2qy + r = 0$



8. **(c)** $t_{r+1} = \dfrac{x}{(rx+1)\{(r+1)x+1\}} = \dfrac{(r+1)x+1-(rx+1)}{(rx+1)[(r+1)x+1]} = \dfrac{1}{(rx+1)} - \dfrac{1}{(r+1)x+1}$

$\therefore \quad S_n = \sum\limits_{r=0}^{n-1} t_{r+1} = \begin{cases} 1 - \dfrac{1}{nx+1}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

$\therefore \quad f(x) = \lim\limits_{n\to\infty} S_n = \begin{cases} 1, & x \neq 0 \\ 0, & x = 0 \end{cases} \qquad \therefore \quad \lim\limits_{x\to 0} f(x) = 1$ and $f(0) = 0$

Hence $f(x)$ is neither continuous nor differentiable at $x = 0$.

Clearly $f(x)$ is not a periodic function

SECTION – II
Multiple Correct Choice Type

9. (a), (b)

$$\arg\frac{z_1}{z_2}=\pm\pi \text{ and } |z_1+i(z_3-z_1)|=|z_1|+|z_3-z_1| \text{ iff } \arg\frac{z_1}{z_3-z_1}=\frac{\pi}{2}$$

So centre of circle = $\left(\frac{z_2+z_3}{2}\right)$ and ABC is right angle triangle.

10. (c), (d)

$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}=2R$$

$da = 2R\cos A\, dA$, $db = 2R\cos B\, dB$,

$dc = 2R\cos C\, dC$

$$\frac{da}{\cos A}+\frac{db}{\cos B}+\frac{dc}{\cos C}=2R\,(dA+dB+dC)$$

also $A + B + C = \pi$

$\therefore$ $dA + dB + dC = 0$.

11. (c), (d)

$$\lim_{n\to\infty}\tan\left(\frac{1}{n}\right)\ln\left(\frac{1}{n}\right)=0 \qquad \text{[By L' Hospital's rule]}$$

$$\Rightarrow \quad f(x)=1$$

$$=\int\frac{dx}{\sin^{11/3}x\cos^{1/3}x}=\int\frac{\sec^4 x}{\tan^{11/3}x}dx=\int\frac{(1+\tan^2 x)\sec^2 x}{\tan^{11/3}x}dx$$

$$=\frac{-3}{8}(\tan x)^{-8/3}-\frac{3}{2}(\tan x)^{-2/3}+C$$

$$g\left(\frac{\pi}{4}\right)=-\frac{15}{8}.$$

12. (d)

Let O be the origin $\overrightarrow{OA}=\hat{i}+\hat{j},\overrightarrow{OB}=\hat{i}-\hat{j},\overrightarrow{OC}=p\hat{i}+q\hat{j}+r\hat{k}$

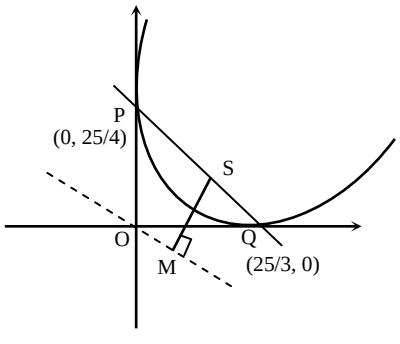
$\overrightarrow{AB}=-2\hat{j},\ \overrightarrow{AC}=(p-1)\hat{i}+(q-1)\hat{j}+r\hat{k}$

A, B, C are collinear

$\therefore$ $\overrightarrow{AC}=\lambda\overrightarrow{AB}$.

SECTION–III
Comprehension Type

Sol.: Here OP and OQ are perpendicular tangents intersecting at origin we know that
(i) Point of intersection of perpendicular tangents lies on directrix of parabola (shown by doted lines and
(ii) Portion of tangents intercepted between the directrix and point of contact subtend right angle at focus.



Let focus is $S(h, k)$

So $\dfrac{k-\dfrac{25}{4}}{h}\times\dfrac{k}{h}=-1$ and $\dfrac{k}{h-\dfrac{25}{3}}\times\dfrac{k}{h}=-1 \Rightarrow h=3,\ k=4$

$PS=\dfrac{15}{4},\ QS=\dfrac{20}{3}$

If L = latus rectum

$\dfrac{4}{L}=\dfrac{1}{PS}+\dfrac{1}{QS} \Rightarrow \dfrac{4}{L}=\dfrac{4}{15}+\dfrac{3}{20} \Rightarrow L=\dfrac{240}{25}=\dfrac{48}{5}$

Circle passing through P, Q and O is $\left(x-\dfrac{25}{3}\right).x+y.\left(y-\dfrac{25}{4}\right)=0$

Equation of directrix is tangent to above circle at (0, 0).
Equation of directrix is $4x+3y=0$.

13. **(c)**
14. **(c)**
15. **(a)**

SECTION–IV
Matrix–Match Type

1. **A-1, 4** Clearly $\lim\limits_{x\to\frac{\pi}{2}}\left[\dfrac{|\sin x|-|\cos x|+1}{3}\right]=\left[\dfrac{2}{3}\right]=0$

B-2, 4 $f'(x+y)=f(x).f'(y)$ (keeping x as a constant)
$\Rightarrow f'(x)=2f(x) \Rightarrow \log f(x)=2x+c$
$\Rightarrow f(x)=ce^{2x} \Rightarrow f(0)=1 \Rightarrow c=1$

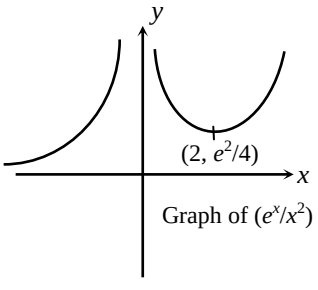
So $f(x)=e^{2x}$

$f\left(\frac{x}{2}\right)=e^{x}$

Now $\frac{e^{x}}{x^{2}}=k$

For $x>0$ minimum value of $\frac{e^{x}}{x^{2}}$ is $\frac{e^{2}}{4}$

$\Rightarrow$ For exactly one solution $0<k<\frac{e^{2}}{4}$.



C-2, 3, 4 $\int_{-2}^{-1}1\,dx+\int_{-1}^{1}0\,dx+\int_{1}^{2}1\,dx=2$

D-1, 4 $g(x)=\sin^{-1}([x]+1)+\cos^{-1}[x]$, domain $-1\le x<1$

Range of $g(x)=\pi$

$f(x)+g(x)=3$

$\cos^{-1}(\cos x)=3-\pi$

This is not possible, hence no solution.

2. (A)-3, 4 Hence $(1+\tan k)(1+\tan(45-k))=(1+\tan k)\left(1+\frac{1-\tan k}{1+\tan k}\right)$

It follows that

$(1+\tan 1°)(1+\tan 2°)...(1+\tan 45°)$

$=(1+\tan 1°)(1+\tan 44°)(1+\tan 2°)(1+\tan 43°)$

$\ldots(1+\tan 22°)(1+\tan 23°)(1+\tan 45°)$

$=2^{23}$.

$\Rightarrow \lambda+1=24$

(B)-4 Solving $x+2y+3z=7,\ 2x-y+z=5$ we get $x-3y-2z=-2$

So if $\lambda = 2$, then

$x-3y-2z=-2$ and $x-3y-2z=1$ are in-consistent.

(C)-3, 5 Clearly the line is parallel to the plane.

If $(1,-|\lambda|,\ 0)$ lies on the plane $2x-y-z=5$

$2+|\lambda|-0=5 \Rightarrow \lambda=\pm 3$

(D)-1, 4 $A(1,\ 0,\ 0),\ B(0,\ a,\ 0),\ C(0,\ 0,\ a)$

Area of $\Delta ABC=\frac{1}{2}|a|.\sqrt{2+a^{2}}=\sqrt{6} \Rightarrow a=\pm 2$

SECTION–V
Integer Answer Type

1. Replace x by $-x$

$$2f(-x)+f(x)=\frac{1}{x}\sin\left(x-\frac{1}{x}\right)$$

and solving with the given equation, we get $3f(x)=\frac{1}{x}\sin\left(x-\frac{1}{x}\right)$

$$\Rightarrow I=\frac{1}{3}\int_{1/e}^{e}\frac{1}{x}\sin\left(x-\frac{1}{x}\right)dx \qquad \text{put } x=\frac{1}{t}$$

$$I=\frac{1}{3}\int_{e}^{1/e}t\sin\left(\frac{1}{t}-t\right)\left(-\frac{1}{t^2}\right)dt=-\frac{1}{3}\int_{1/e}^{e}\frac{1}{t}\sin\left(t-\frac{1}{t}\right)dt=-I$$

$$\Rightarrow I=0$$

Ans. 0

2. Probability of hitting the target in one fire $p=\frac{1}{5}$

Then probability of hitting the target at least once in n fires

$=1-$(probability of not hitting the target)

$$=1-\left(\frac{4}{5}\right)^n>\frac{3}{4}$$

$$\Rightarrow \left(\frac{4}{5}\right)^n<\frac{1}{4} \text{ as } \left(\frac{4}{5}\right)^6>\frac{1}{4} \text{ and } \left(\frac{4}{5}\right)^7<\frac{1}{4}$$

The least value of $n = 7$

Ans. 7

3. $\sum_{r=0}^{200}\alpha_r x^r=\sum_{r=0}^{200}\beta_r(1+x)^r$

$\alpha_0+\alpha_1 x+\alpha_2 x^2+.....+\alpha_{200}x^{200}$

$=\beta_0+\beta_1(1+x)+.....+\beta_{200}(1+x)^{200}$

Equating coefficient of x^{100}, we get $\alpha_{100}={}^{100}C_{100}+{}^{101}C_{100}+.....{}^{200}C_{100}={}^{201}C_{101}$

Similarly we can find $\alpha_{101},.....\alpha_{200}$.

$\sum_{r=100}^{200}\alpha_r={}^{201}C_{101}+{}^{201}C_{102}+.....{}^{201}C_{201}$

$A=2^{200}$

When A is divided by 15 remainder is 1.

Ans. 1

4. In the first round all integers, which leave remainder 1 when divided by 15, will be marked,

last number of this category is 991. Next number will be 991 + 15 – 1000 = 6. That means in second round all integers, which leave remainder 6 when divided by 15 will be marked. In short numbers of the form $5k + 1$ will be marked.

$\therefore \quad A = 200$ and $\frac{A}{50} = 4$.

Ans. 4

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